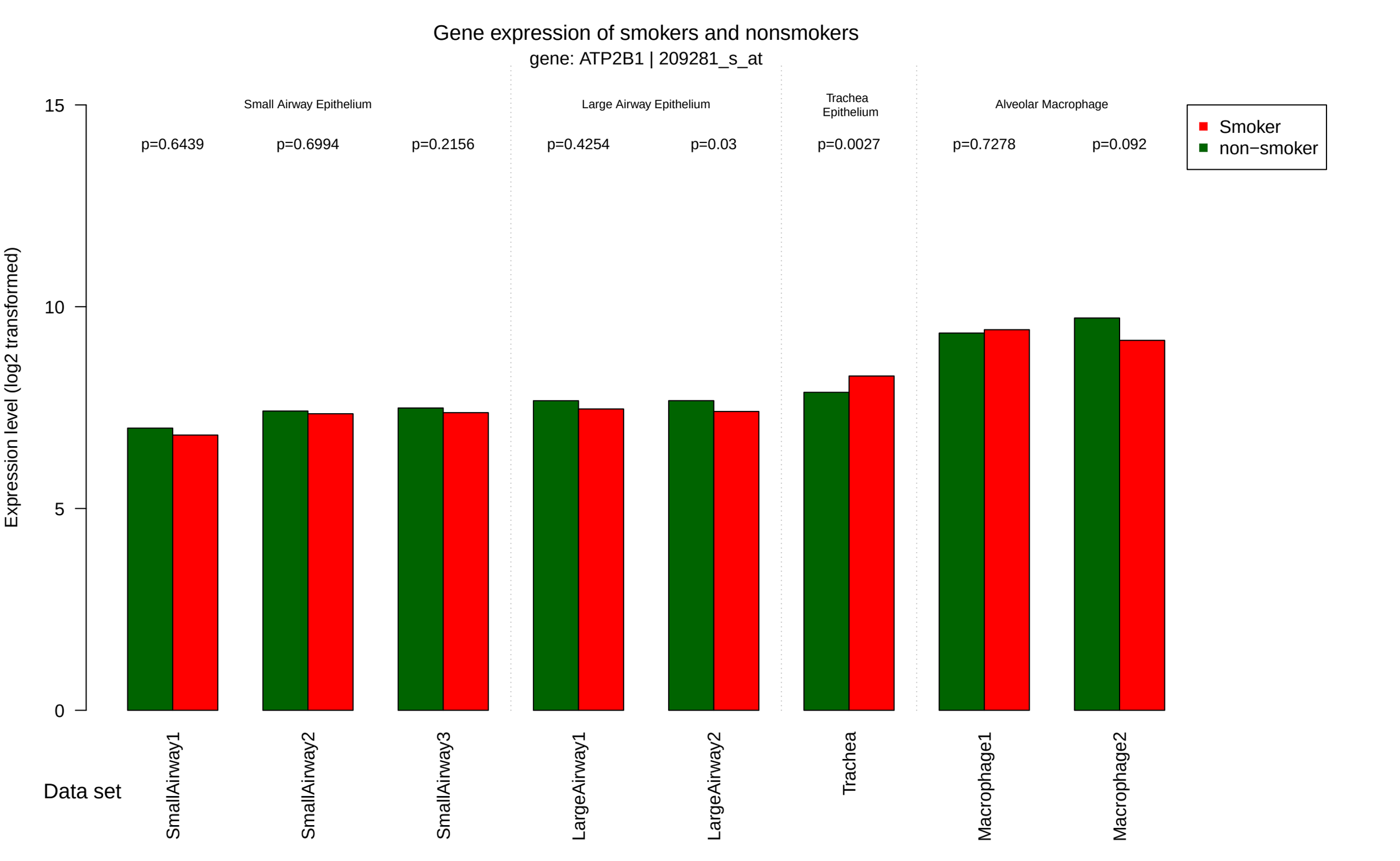
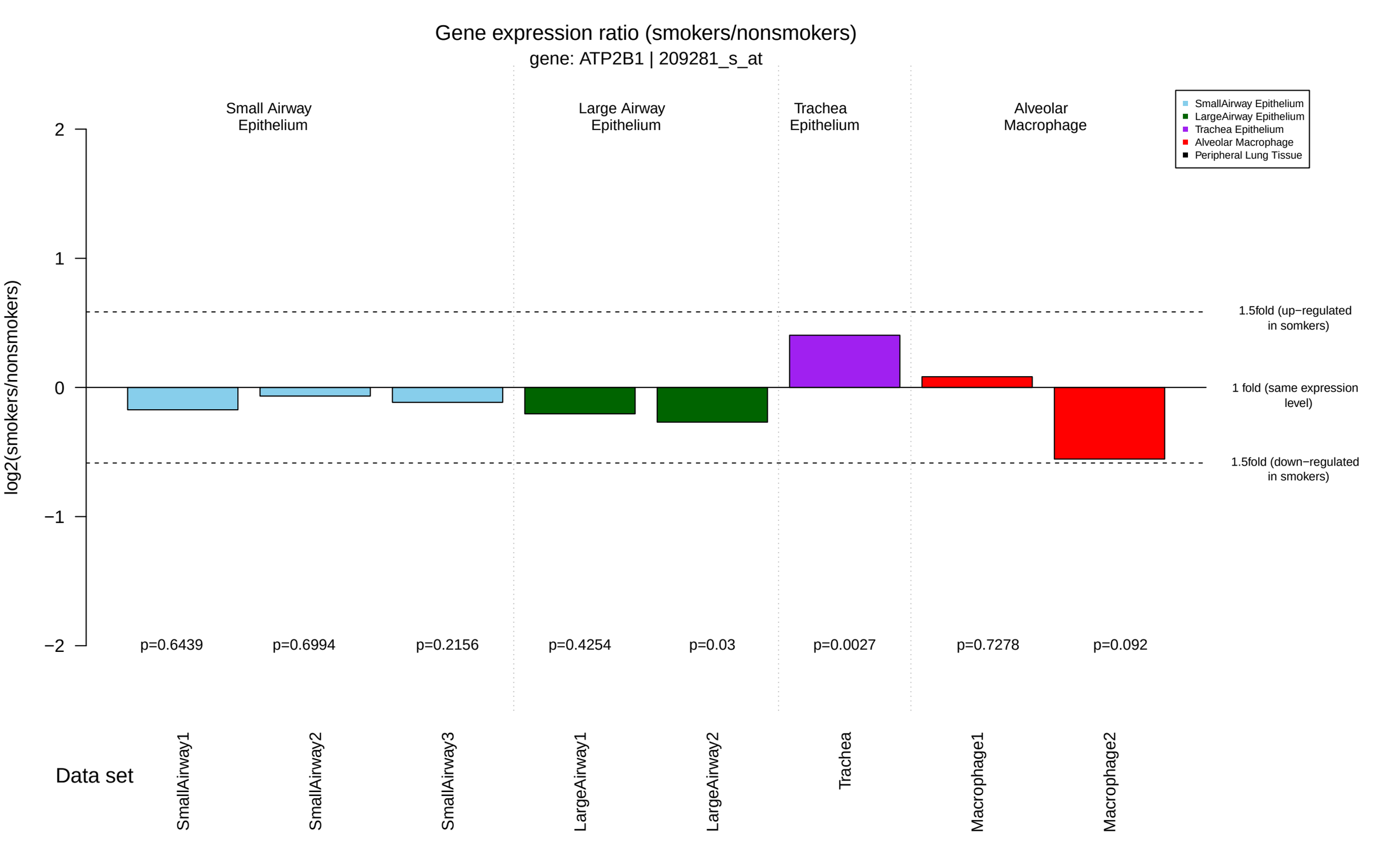
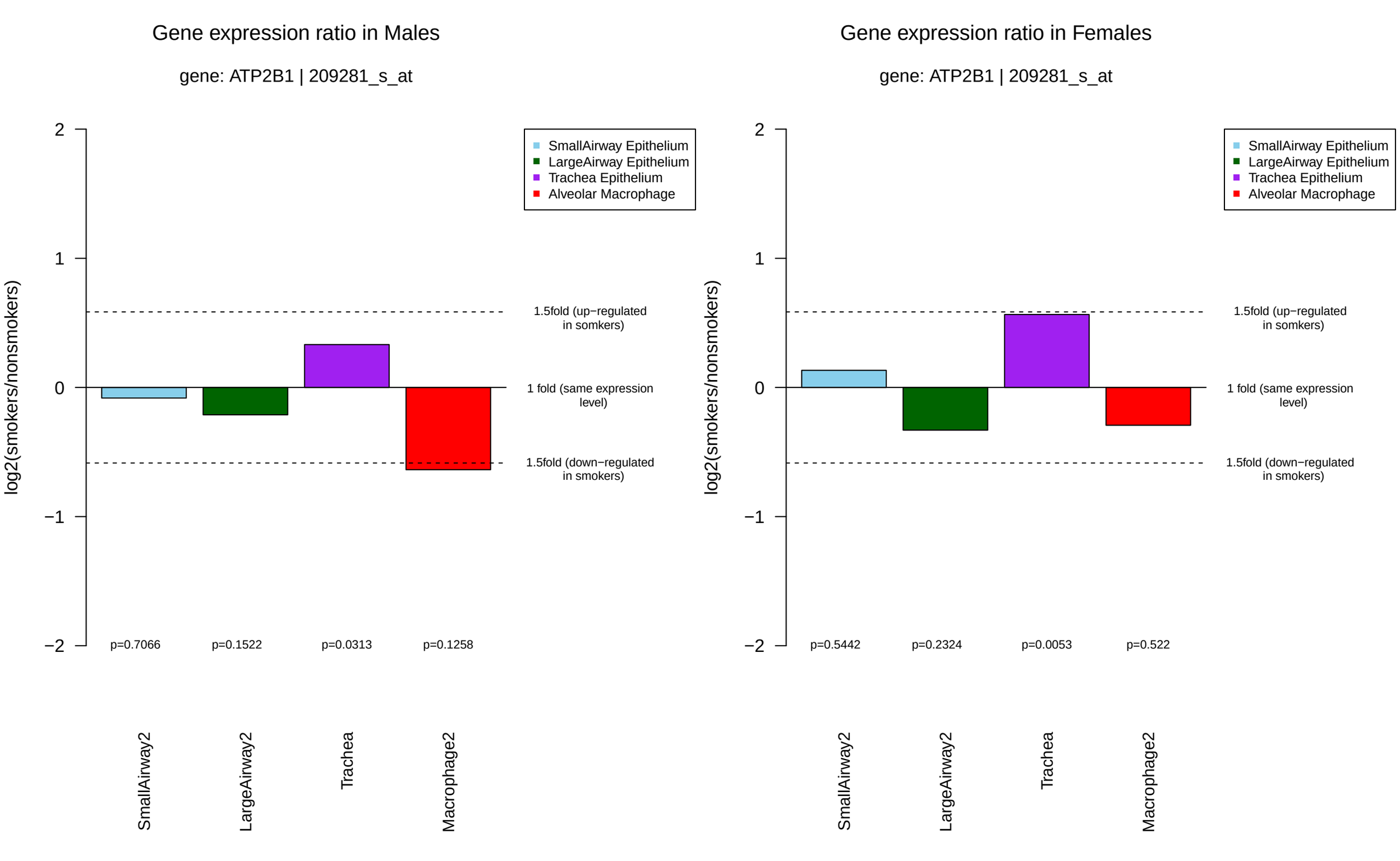
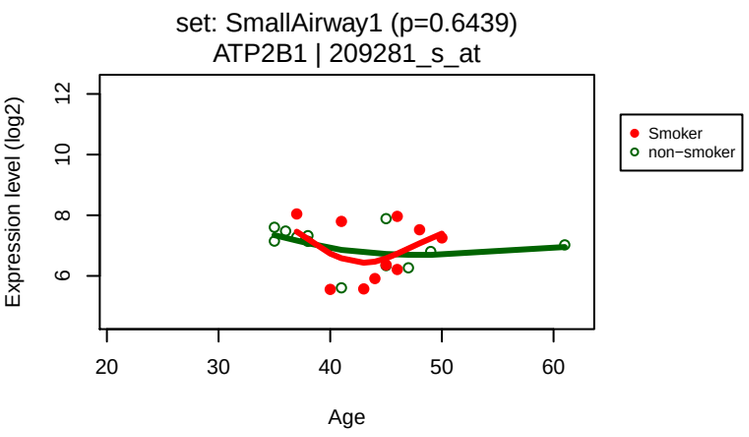
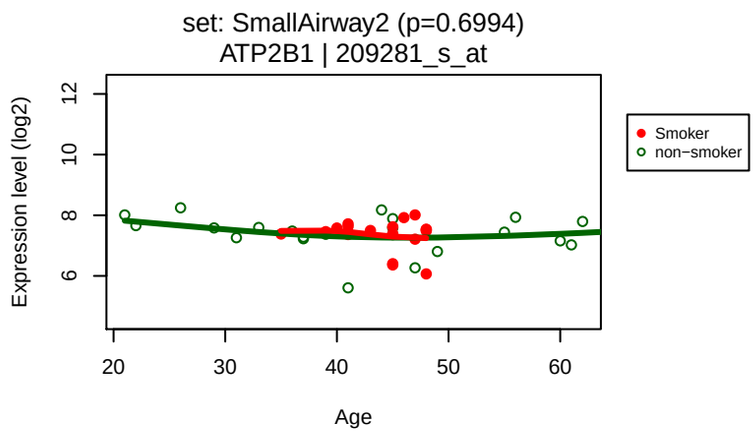
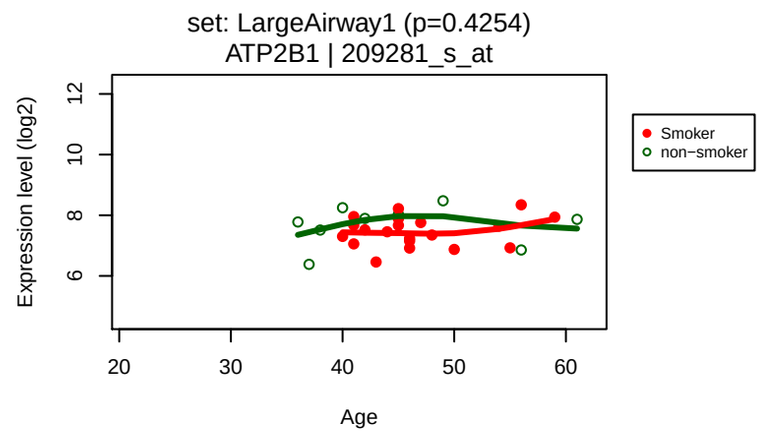

Gene expression of smokers and nonsmokers
gene: ATP2B1 | 209281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6439
p=0.6994
p=0.2156
p=0.4254
p=0.03
p=0.0027
p=0.7278
p=0.092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

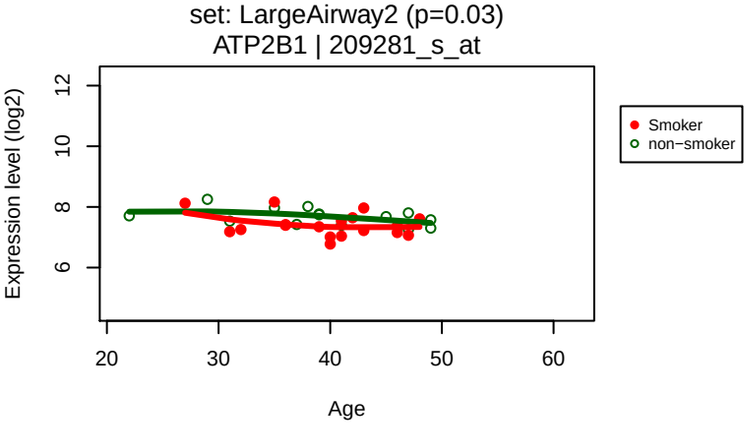

Gene expression ratio (smokers/nonsmokers)
gene: ATP2B1 | 209281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6439
p=0.6994
p=0.2156
p=0.4254
p=0.03
p=0.0027
p=0.7278
p=0.092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

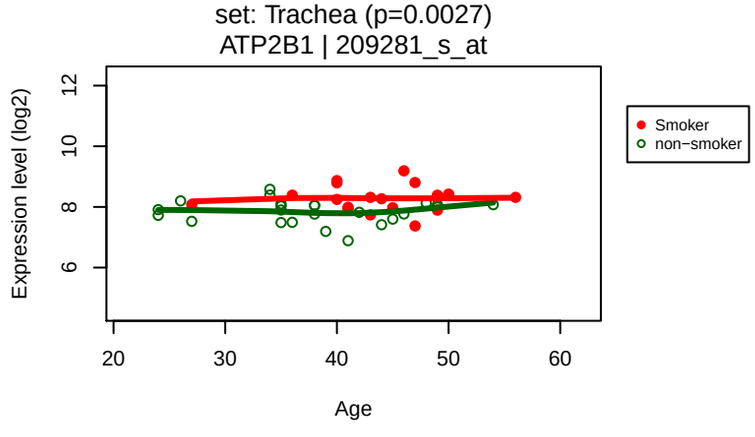

Gene expression ratio in Males
gene: ATP2B1 | 209281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7066
p=0.1522
p=0.0313
p=0.1258
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2B1 | 209281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5442
p=0.2324
p=0.0053
p=0.522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

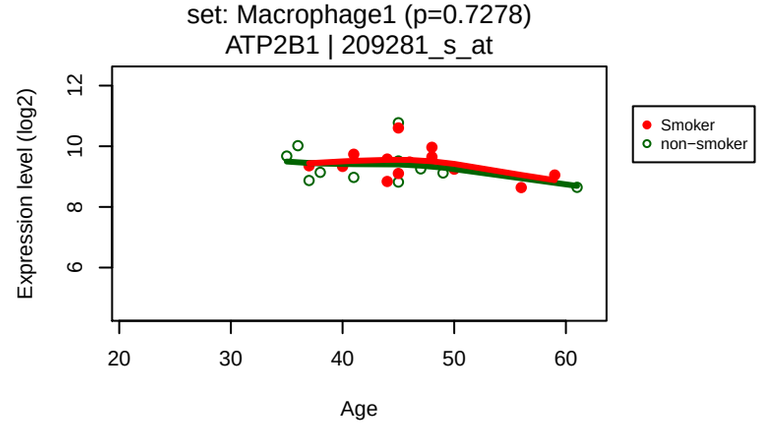

set: SmallAirway1 (p=0.6439)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker

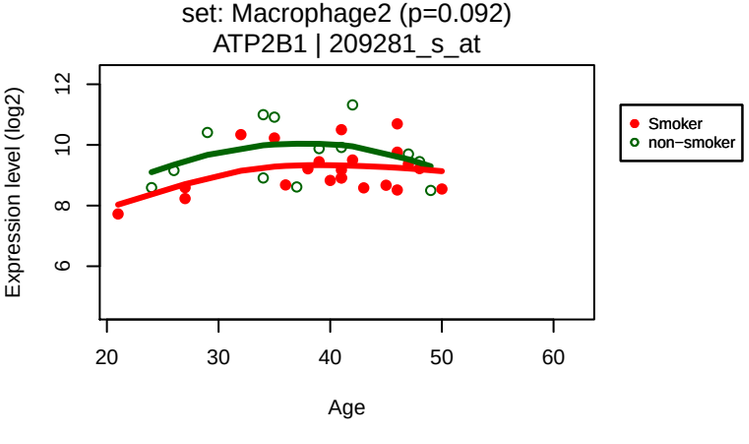

set: SmallAirway2 (p=0.6994)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker

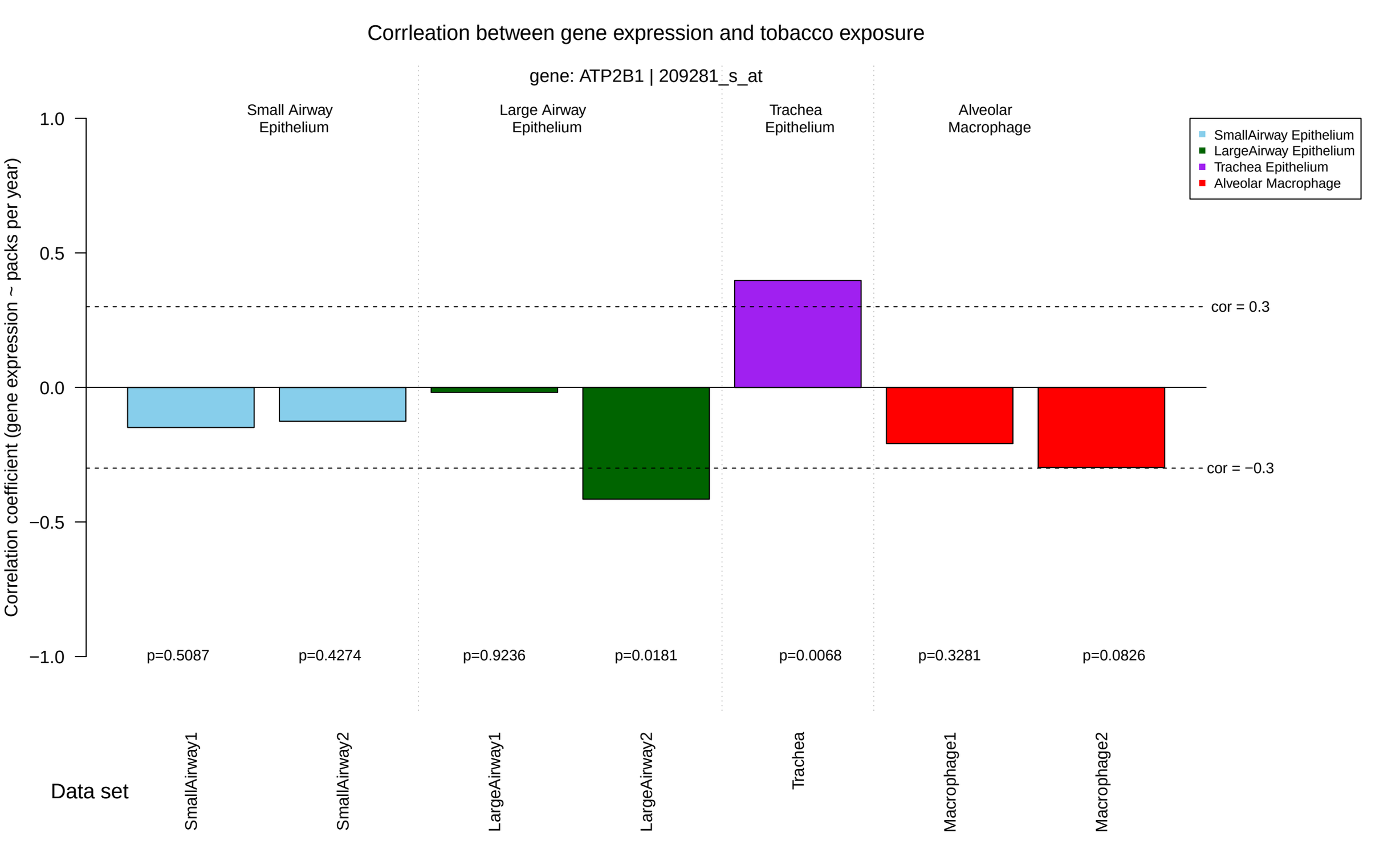

set: LargeAirway1 (p=0.4254)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.03)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0027)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7278)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.092)
ATP2B1 | 209281_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2B1 | 209281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5087
p=0.4274
p=0.9236
p=0.0181
p=0.0068
p=0.3281
p=0.0826
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2